

Coordinates

Suppose that $\mathcal{B} = \{\underline{v}_1, \dots, \underline{v}_n\}$ is a basis for a vector space V . Then any $v \in V$ can be written uniquely as

$$\underline{v} = \underline{c_1 v_1 + \dots + c_n v_n}$$

for some sequence of numbers $c_1, \dots, c_n$. We call this tuple the coordinates of v with respect to $\mathcal{B}$ and we write

$$[v]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$$

Ex: P_2 - polys of deg. ≤ 2 .

$\mathcal{B}_1 = \{\underline{1}, \underline{x}, \underline{x^2}\}$ is a basis for P_2

$\mathcal{B}_2 = \{\underline{1}, \underline{(1+x)}, \underline{(1+x+x^2)}\}$ is also a basis for P_2 .

Ex: What are the coordinates of x^2 wrt B_1 and B_2 ?

$$[x^2]_{B_1} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^3$$

$$[x^2]_{B_2} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \in \mathbb{R}^3 \quad \text{since}$$
$$x^2 = (1+x+x^2) - (1+x)$$

Coordinates as a linear transformation

Suppose that $\underline{B} = \{v_1, \dots, v_n\}$ is a basis for V . Then the map $T: V \rightarrow \mathbb{R}^n$ given by

$$T(v) = [v]_B$$

is a linear transformation which is one-to-one and onto. Such linear transformations are called isomorphisms.

If V has a basis with n vectors then as far as vector spaces are concerned, V and $\mathbb{R}^n$ are the same (isomorphic).

Ex: $V = P_2$ with $B = \{1, x, x^2\}$ as a basis then

$T(v) = [v]_B \in \mathbb{R}^3$ so P_2 and $\mathbb{R}^3$ are isomorphic.

In general, P_n has a basis $\underbrace{\{1, x, x^2, \dots, x^n\}}_{n+1 \text{ vectors}}$

so P_n is isomorphic to $\mathbb{R}^{n+1}$

Why is T a lin. trans? For V with a basis, $v_1, \dots, v_n$.

$$T(v) = [v]_{\mathcal{B}}$$

$$v, w \in V$$

$$v = c_1 v_1 + \dots + c_n v_n$$

$$w = d_1 v_1 + \dots + d_n v_n$$

$$v+w = \underbrace{(c_1+d_1)}_{v_1} + \dots + \underbrace{(c_n+d_n)}_{v_n} \quad [w]_{\mathcal{B}} = \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix}$$

$$[v]_{\mathcal{B}} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$$

$$\begin{aligned}
 T(v+w) &= \begin{pmatrix} c_1 + d_1 \\ \vdots \\ c_n + d_n \end{pmatrix} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} + \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix} \\
 &= T(v) + T(w)
 \end{aligned}$$

Dimension

Suppose that V has a finite spanning set. Then every basis of V is finite and has the same size. The size of any such basis is called the dimension of V written $\dim(V)$. We say that V is finite dimensional. If V does not have a finite spanning set then we call it infinite dimensional.

Ex: ① $\mathbb{R}^n$ has dimension n . $\leftarrow$

② P_n has dim. $n+1$

③ V is the set of $m \times n$ matrices.

What is $\dim(V)$? $e_{ij} = \begin{cases} 1 & \text{in the } i^{\text{th}} \text{ spot} \\ 0 & \text{otherwise.} \end{cases}$
 $1 \leq i \leq m, 1 \leq j \leq n.$

is a basis for V . $\dim(V) = mn.$

④ $V = \text{all } n \times n \text{ skew-symmetric matrices.}$

All $A, n \times n$, s.t. $\underline{A^T} = \underline{-A}$.

A skew-symmetric matrix looks like

$$\begin{pmatrix} 0 & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{pmatrix}$$

What could be a basis?

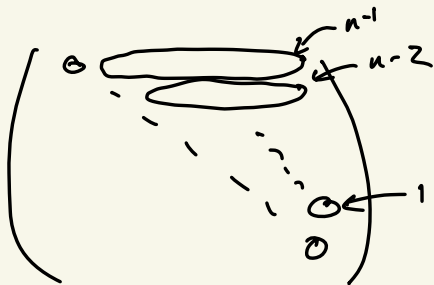
$$e_{ij} - e_{ji} \quad 1 \leq i < j \leq n$$

3x3 skew-sym:

3 vectors form
a basis.

$$\rightarrow \underline{\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}, \underline{\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}}, \underline{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}}$$

The dim in the $n \times n$ case is



$\frac{n(n-1)}{2}$ many
basis vectors.

$$\dim(V) = \frac{n(n-1)}{2}$$

Check: If V is all $n \times n$ sym. matrices then
$$\dim(V) = \frac{n(n+1)}{2}$$

Ex: P , all poly. has inf. dimension.

$\{1, x, x^2, x^3, x^4, \dots\}$ is a basis for P .

Why is P not Span $\{p_1, \dots, p_n\}$ for some
poly. $p_1, \dots, p_n$? We can't generate any
 x^m where $m > \deg(p_1), \deg(p_2) \dots \deg(p_n)$.

$P \subseteq$ all cont. functions on $\mathbb{R}$.

Subspaces

Suppose that $n = \dim(V)$.

- 1 Then every set of vectors of size greater than n is linearly dependent,
- 2 no set of vectors of size less than n spans V ,
- 3 every linearly independent set of size n is a basis of V ,
- 4 every spanning set of size n is a basis of V , and
- 5 if H is a subspace of V then $\dim(H) \leq n$.

- ① Suppose not. Say $v_1 \dots v_k$ are lin indep and $k > n$.
Extend $v_1 \dots v_k$ to a basis of V but then V
has a basis of size $> n$.
- ② If $v_1 \dots v_k$ spans V and $k < n$ then V has
a basis contained in $v_1 \dots v_k$ which has size $< n$.

Rank and nullity

If A is an $m \times n$ matrix then the rank of A is the dimension of the column space of A . The nullity of A is the dimension of the nullspace of A .

Theorem

For an $m \times n$ matrix A , then

$$\text{rank}(A) + \text{nullity}(A) = n.$$

Moreover, the rank of A is the dimension of the row space of A .